



PROJECT-BASED LEARNING

PHASE-I EVALUATION

**Intelligent Emergency Response
and Traffic Network Optimization
for Smart Cities**

Team ID: DSCPP-III-2026-T230

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T230
Semester: 3rd Semester
Section: A, G, E
Domain: Smart City / Transportation

Team Members

Harsh Mittal — 2510011830
Dhairya Saini — 2510011061
Dhruv Kstharia — 2510019427

Mentor

Mr. Rohan Verma

Problem Statement & Motivation

Problem Statement

Traffic congestion and red signals can delay emergency vehicles and increase response time.

Why is it a problem?

- ▶ Congested roads cause delays.
- ▶ Signal stops increase travel time.
- ▶ Emergencies require quick response.

Our Motivation

- ▶ Find efficient emergency routes.
- ▶ Reduce waiting at signals.
- ▶ Improve emergency response.

Focus: Software-based emergency response in a simulated city.

Objectives

Main Objectives

1. Model the city road network as a graph.
2. Find efficient routes for emergency vehicles.
3. Provide priority at selected intersections.
4. Simulate different traffic situations.
5. Compare response time and travel efficiency.

Key Objective

Combine route optimization with traffic signal priority to reduce emergency vehicle delay.

Proposed Solution

How it works

1. Create a graph of the road network.
2. Enter emergency vehicle source and destination.
3. Find an efficient route.
4. Identify important intersections.
5. Give priority at selected signals.
6. Evaluate the response.

Key Features

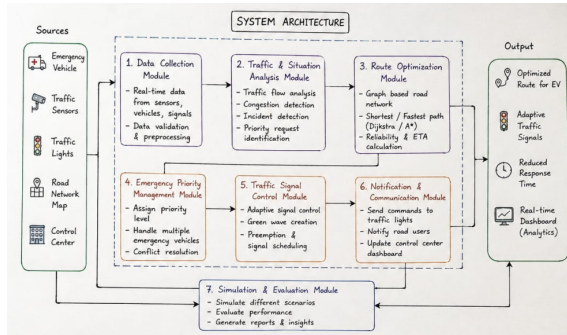
- ▶ Graph-based road network
- ▶ Emergency route selection
- ▶ Traffic signal priority
- ▶ Traffic simulation
- ▶ Response-time comparison
- ▶ No physical hardware

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Arrays, Linked Lists, Trees, Graphs, Hashing, etc.	Road Network Module
OOPs with C++	Classes, Inheritance, Polymorphism, STL, etc.	Emergency Vehicle Module

Requirement: Demonstrate meaningful integration of both subjects in **one** project.

System Architecture / Workflow



Emergency Input → Road Network → Route Optimization → Traffic Priority → Response

Technology Stack & Methodology

Technology

- ▶ C++ / Python
- ▶ Graph-based data structures
- ▶ Route optimization algorithms
- ▶ Software-based simulation
- ▶ Visual Studio Code
- ▶ Git / GitHub

Methodology

1. Requirement Analysis
2. Literature Study
3. System Design
4. Network Modelling
5. Algorithm Implementation
6. Testing & Evaluation

The initial implementation is software-based and does not require physical hardware.

Initial Progress & Team Contribution

Progress

- ▶ Problem identified
- ▶ Literature study completed
- ▶ Existing approaches reviewed
- ▶ Proposed approach defined
- ▶ Architecture prepared
- ▶ Future milestones identified

Team Contribution

Member	Role	Contribution
Harsh	Team Lead	Planning, literature study and system design
Dhairya	Research	Literature review and problem analysis
Dhruv	Development	Architecture, documentation and technical planning

Roadmap, Outcomes & References

Roadmap

1. Phase-I: Design
2. Phase-II: Development
3. Phase-III: Testing
4. Future: Real-time extensions

Expected Outcomes

- ▶ Emergency response prototype
- ▶ Route optimization
- ▶ Traffic priority
- ▶ Simulation results

References

1. Dodia et al., "EVATL: A novel framework for emergency vehicle communication with adaptive traffic lights," IET Smart Cities, 2023.
2. Relevant literature on emergency vehicle priority and adaptive traffic signal control.
3. Dijkstra, "A Note on Two Problems in Connexion with Graphs," 1959.
4. Federal Highway Administration, "Traffic Signal Timing Manual."

Thank You
Questions & Suggestions